



VoltRescue

Business process and operations

Document	VoltRescue Process Flow Guide
Phase	POC — Dar es Salaam pilot, Phase 1
Prepared for	Mr. Menelick Erick, COO
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VoltRescue — Business Process Flow & Flowchart Guide

Document 1 of 3 · Business process and operations **Audience:** Supervisor (Mr. Menelick Erick), operations staff, product owners, recycling partners, municipal stakeholders **System version:** VoltRescue POC — Dar es Salaam pilot, Phase 1 **Application URL (local):** <http://127.0.0.1:8815/> **Source of truth:** SQLite database `data/voltrescue.sqlite`, reached only through the REST API **Companion documents:** `02_VoltRescue_User_Guide.md` (how to use it), `03_VoltRescue_Technical_Design_Handbook.md` (how it is built)

Reading note. Diagrams in this file use Mermaid. If your Markdown preview shows a resource error, open the file as plain text; the diagram source is still readable line by line.

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1. Solution Overview

1.1 What VoltRescue is

VoltRescue is a **battery reverse-logistics platform**. It moves spent batteries from the household or small business that owns them, through a named collector, into the hands of a licensed recycling partner — and records every step of that journey as a permanent database record.

The Phase 1 POC is a single web application. Citizens, collectors, recycler staff, and administrators all open the same address and log in; the menu and dashboard they see are decided by the role stored against their account. The interface is dark-themed and mobile-first, because collectors work from phones in the field.

Four things make it a platform rather than a form:

- **One request record.** Every pickup exists once, with an ID, a GPS point, a battery type, a quantity, and an owner.
- **A forced lifecycle.** A request cannot jump from "submitted" to "completed". It must pass through each stage, and each stage is stamped with who changed it and when.
- **A chain of custody.** The system knows which collector physically holds the batteries and which recycler received them.
- **An audit trail.** Every login, creation, assignment, status change, upload, and override writes an audit row that an administrator can read later.

1.2 The business problem it solves

Spent lead-acid and lithium batteries in Dar es Salaam are largely collected informally. Coordination happens over phone calls and WhatsApp messages. This creates four operational problems:

Problem	Consequence today	How VoltRescue answers it
No single request record	Requests are lost in chat history; nobody knows the backlog	Every request is a database row with a status and an owner
No proof of custody	If a load disappears between household and recycler, nobody can say who held it	<code>assignments</code> and <code>custody_transfers</code> name the collector and the recycler with timestamps
No location data	Collectors phone the customer repeatedly for directions	GPS coordinates are captured on the map and turned into navigation links
No oversight	Management cannot answer "how many pickups today, how many stuck"	Admin dashboard with live counters, search, filter, and CSV export

1.3 Objectives of the POC

The POC was built to demonstrate, with working software rather than slides, that:

1. A citizen can submit a pickup request that **persists in a database**, not just in a browser tab.
2. An administrator can **assign work** to a named collector and see the whole operation.
3. A collector can **accept, travel, collect, and transfer custody** from a phone.
4. A recycler can **confirm receipt, validate materials, and close** the job.
5. **Illegal status jumps are rejected** by the server unless an administrator deliberately overrides them.
6. Notification, mobile-money, and mapping integrations have **real architectural hooks** — sandbox-ready, with no live money movement.

1.4 Scope covered in Phase 1

Included and working	Excluded from Phase 1 (architecture hooks only)
Registration, login, password reset, profile editing	AI battery image recognition
Pickup request capture with GPS pin and auto-detect	AI scrap-value prediction
Collector assignment and workload visibility	Rewards / points engine
Complete 12-stage pickup and custody lifecycle	Recycling certificates
Recycler receipt, validation, and completion	Carbon credits and sustainability impact reports
Admin KPIs, search, filter, sort, pagination, CSV export	Advanced analytics and commercial BI dashboards
Status override with audit record	Machine learning of any kind
SMS and WhatsApp notification framework (Africa's Talking)	Live mobile-money payouts
M-Pesa sandbox connector — no live debit	Public internet deployment
Evidence photo uploads with preview	Email channel
Full audit logging	

Nothing in the excluded column is faked in the user interface. There are no buttons that pretend to score a battery or issue a certificate.

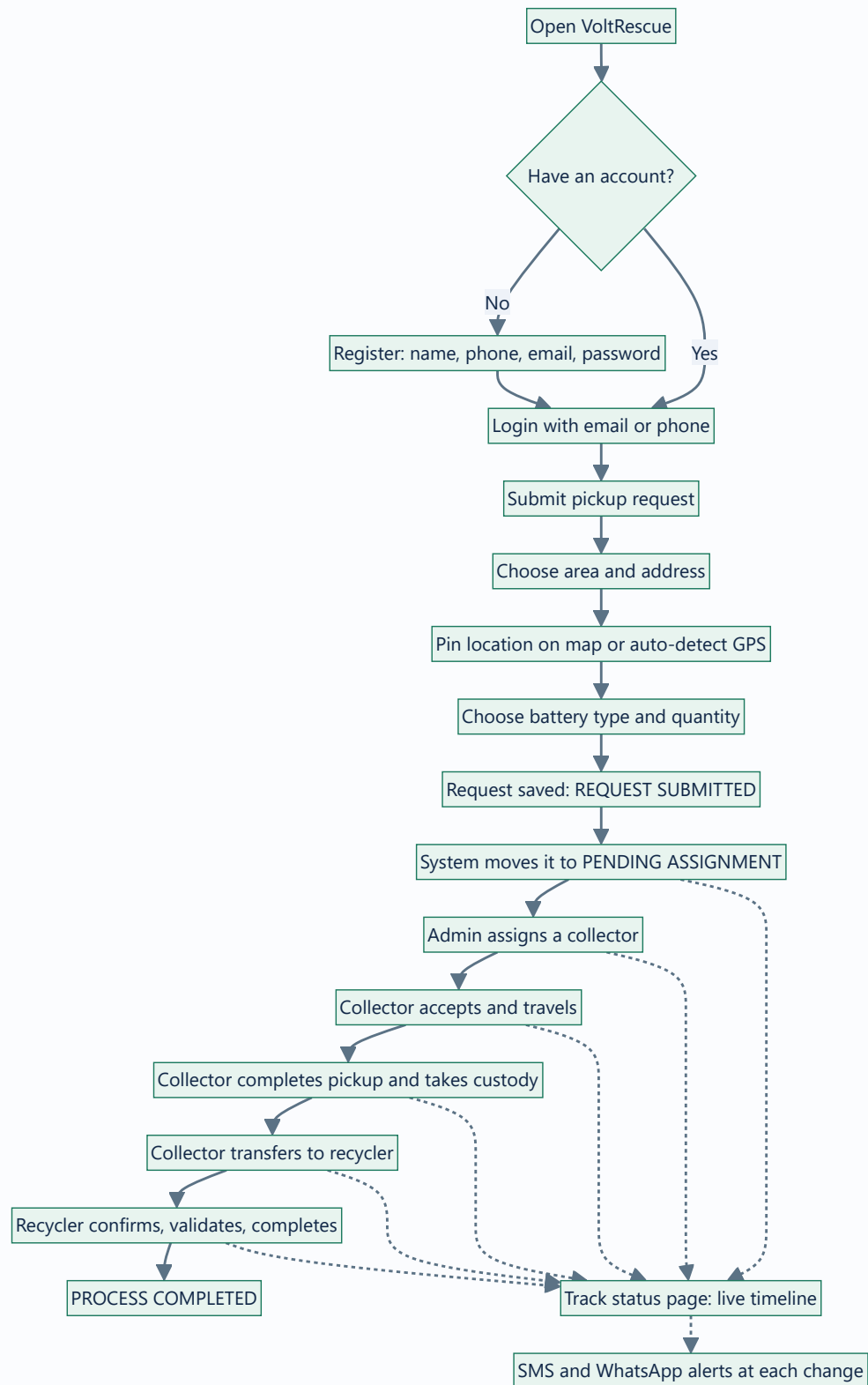
1.5 The four roles at a glance

Role	Who they are	Core responsibility
Citizen / User	Household or small business with spent batteries	Raise the request, track it, receive alerts
Collector	Field agent with a vehicle (bajaj, van)	Take physical custody and deliver to the recycler
Recycler Partner	Licensed processing company (e.g. GreenCycle Dar)	Confirm receipt, validate, process
Administrator	Operations control (Mr. Menelick Erick)	Assign, monitor, override, report

2. Citizen / User Journey

The citizen owns the request. A citizen never assigns a collector and never changes a status beyond cancelling their own request.

2.1 Journey flowchart



2.2 Step-by-step explanation

Step 1 — Registration. The citizen provides name, email, a Tanzanian phone number in the format 255XXXXXXX, and a password of at least eight characters. The account is created with the role `citizen` and the system immediately issues a login token, so registration and first login are one action. A welcome notification is recorded.

Step 2 — Login. The citizen signs in with either email or phone plus password. The server checks the password against a stored hash — it never stores or compares plain text. A successful login writes a `LOGIN` audit entry; a failure writes `LOGIN_FAILED`.

Step 3 — Submit pickup request. The request form asks for area (Kariakoo, Kinondoni, Temeke, Ilala, Ubungu), street address or landmark, battery type (lead-acid, Li-ion, mixed household, automotive), quantity, and optional remarks. A map is shown; the citizen either taps **Use my location** for browser GPS, or clicks / drags the pin. The coordinates are saved with the request.

Step 4 — Automatic queueing. The moment the request is stored it holds the status `REQUEST_SUBMITTED`. The server immediately advances it to `PENDING_ASSIGNMENT` and records both events in the timeline. The citizen is notified, and **every administrator is notified** that a new request needs assignment.

Step 5 — Waiting and tracking. On **Track status** the citizen sees each of their requests as a card with a visual progress bar. Opening a request shows the full timeline: every status, the exact time, the person who caused it, and any note.

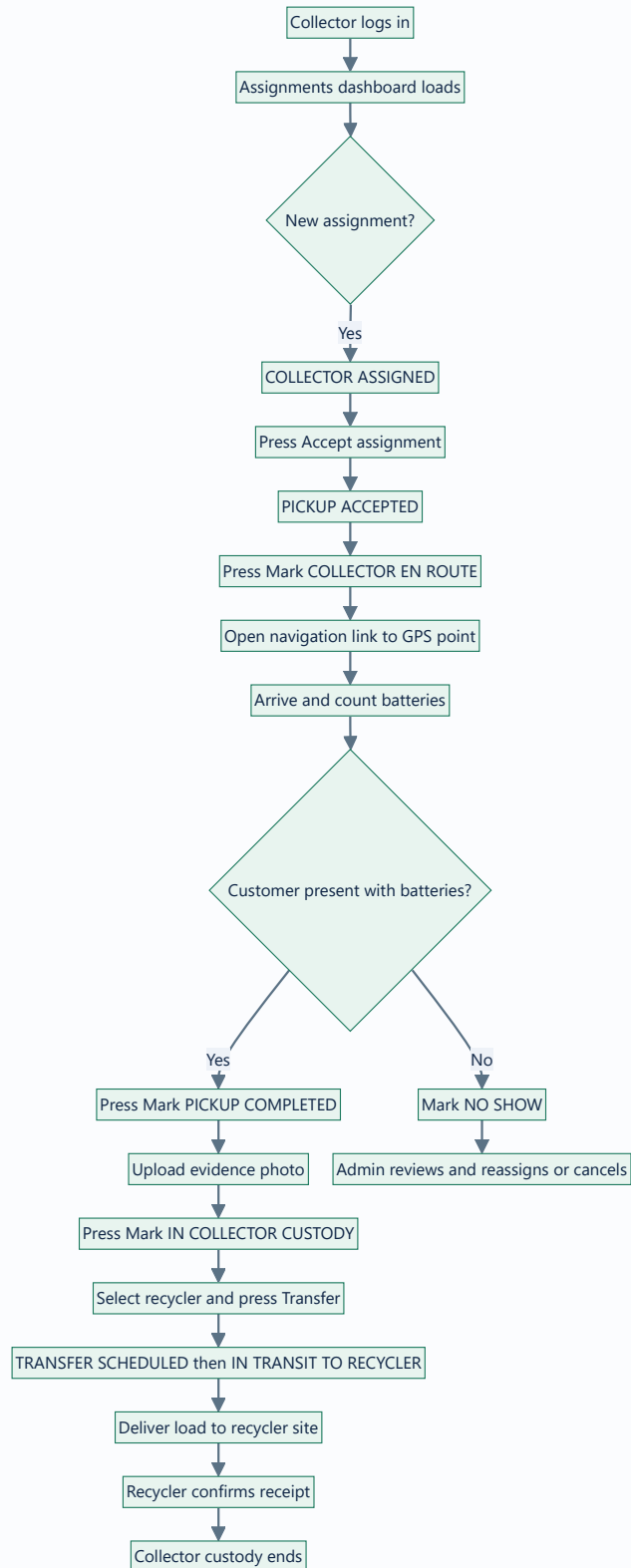
Step 6 — Completion. When the recycler closes the job the request reaches `PROCESS_COMPLETED` and the citizen receives a final alert. The record remains permanently queryable.

What the citizen cannot do: assign collectors, see other people's requests, change status beyond cancellation, view audit logs, or open the admin dashboard. The server enforces this, not just the menu.

3. Collector Journey

The collector is the physical link in the chain and the only role that takes custody of hazardous material.

3.1 Journey flowchart



3.2 Step-by-step explanation

Assignment. The collector does not choose jobs. An administrator assigns a request to a named collector, which sets the status to `COLLECTOR_ASSIGNED` and sends that collector an alert naming the request number and the location.

Accept. The collector presses **Accept assignment**. The server verifies that this request really is assigned to this collector — a collector cannot accept somebody else's job — records the acceptance time on the assignment record, and moves the status to `PICKUP_ACCEPTED`.

Travel. Pressing **Mark COLLECTOR_EN_ROUTE** tells the citizen the collector is on the way. The request card carries a **Navigate** link built from the stored coordinates, opening OpenStreetMap or Google Maps directions.

Collection. After physically loading the batteries the collector presses **Mark PICKUP_COMPLETED**, then **Mark IN_COLLECTOR_CUSTODY**. That second status is the formal statement that the collector is now legally holding the material. Between the two, the collector should upload an evidence photo from the request timeline screen.

Failure at the door. If nobody is there or there is nothing to collect, the collector presses **No-show**. The request moves to `NO_SHOW` and the administrator decides whether to reassign or cancel.

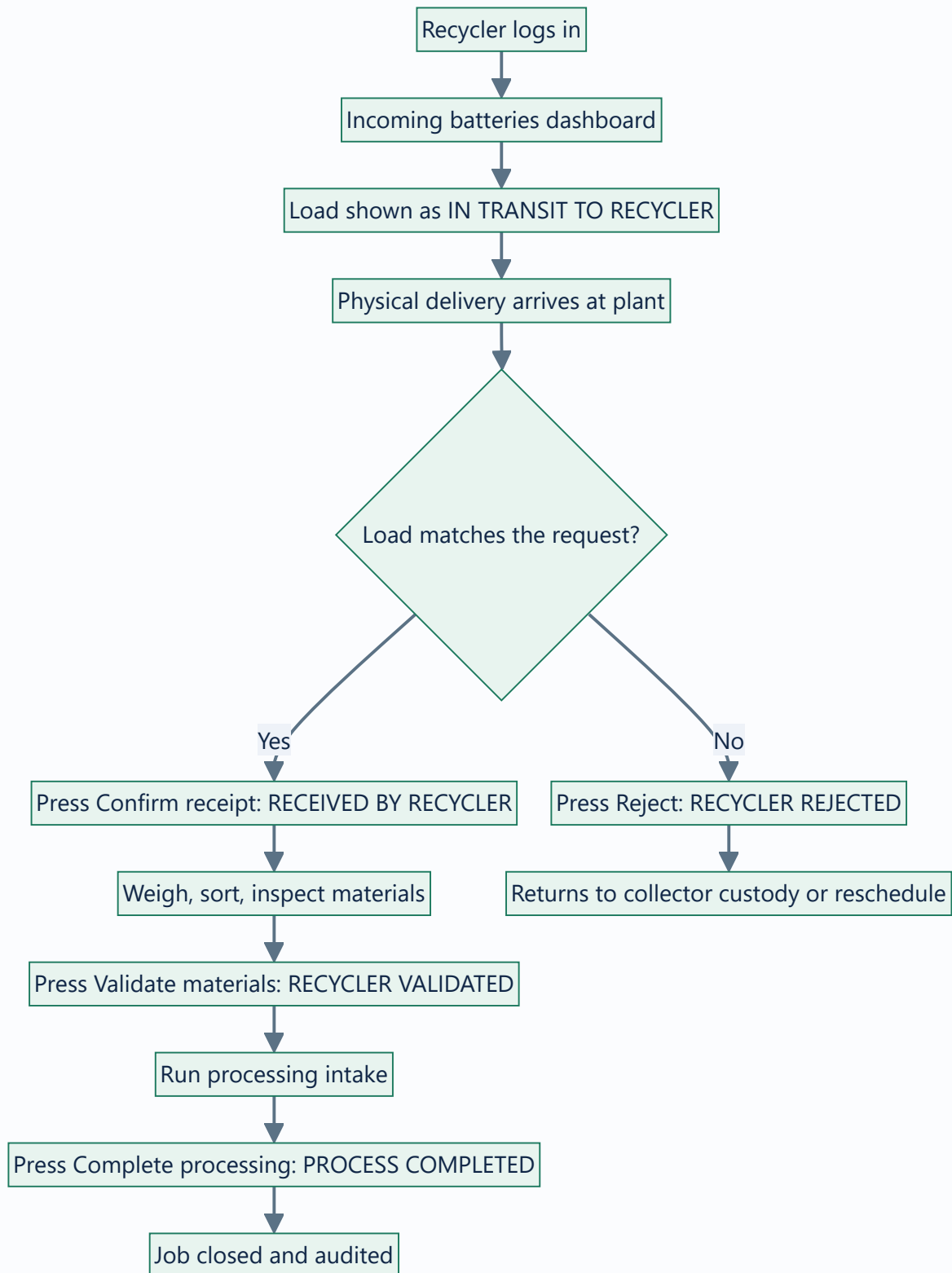
Handover. From a request in custody the collector picks a recycling partner from the dropdown and presses **Transfer to recycler**. This writes a custody transfer record naming the collector, the recycler, and the date, then advances the request through `TRANSFER_SCHEDULED` to `IN_TRANSIT_TO_RECYCLER`. The recycler is notified that a load is inbound.

Release of custody. Custody formally ends only when the recycler confirms receipt. Until then the collector remains the responsible party in the record.

4. Recycler Journey

The recycler is the endpoint of the chain and the only role that can declare a job complete.

4.1 Journey flowchart



4.2 Step-by-step explanation

Visibility. The recycler dashboard lists only requests that concern that recycler — loads in transit, received, validated, rejected, or completed. Requests still sitting with citizens or collectors are invisible to them.

Confirm receipt. When the vehicle arrives, the recycler presses **Confirm receipt**. Status becomes `RECEIVED_BY_RECYCLER`. This is the moment custody legally passes from collector to recycler, and it is timestamped.

Reject. If the load does not match the request — wrong chemistry, damaged cells, quantity mismatch — the recycler presses **Reject**. Status becomes `RECYCLER_REJECTED`, and the request can be sent back into collector custody or rescheduled for another transfer.

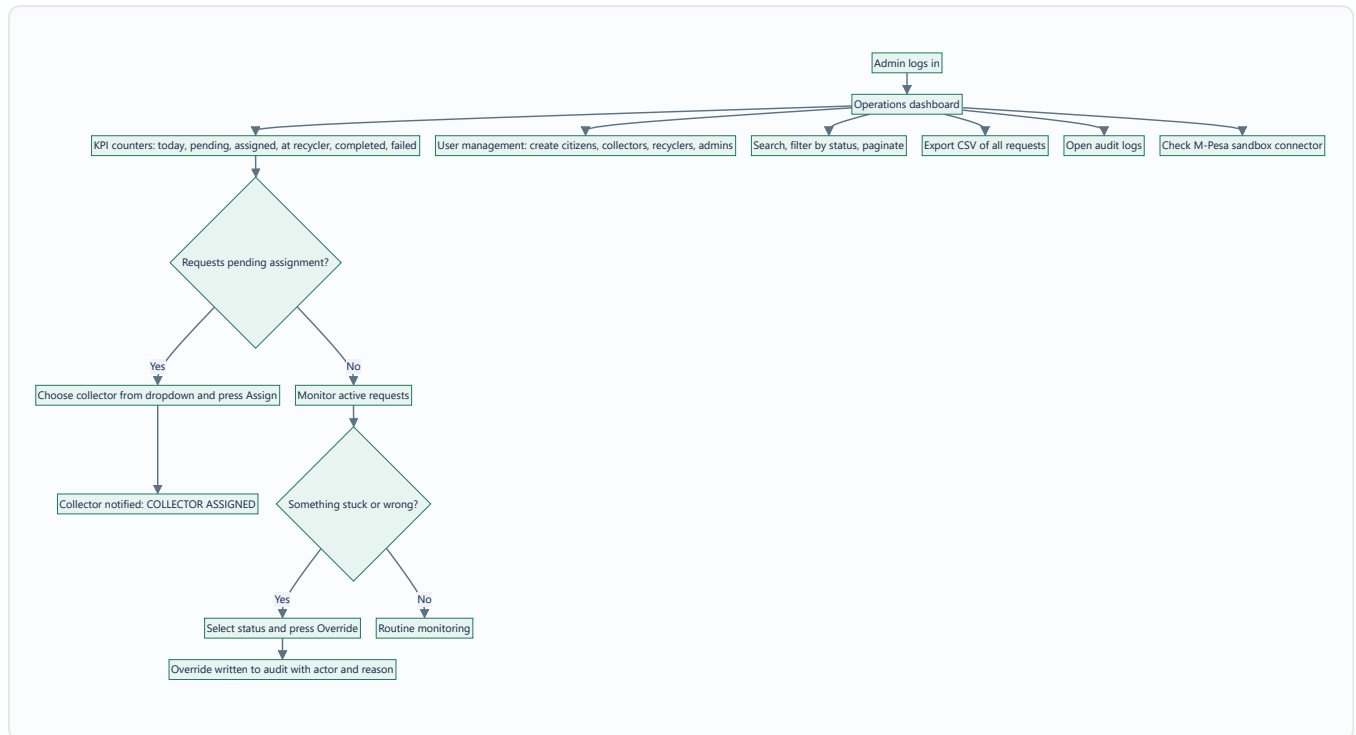
Validate. After weighing and sorting, **Validate materials** sets `RECYCLER_VALIDATED`, confirming the load is genuinely processable.

Complete. Complete processing sets `PROCESS_COMPLETED`, the terminal success state. The citizen receives a final notification. (If a recycler presses Complete directly from "received", the system inserts the validation step automatically so the timeline is never missing a stage.)

5. Admin Journey

The administrator is the operational controller. This is the only role that can see everything and the only role that can break the normal sequence.

5.1 Journey flowchart



5.2 Administrator responsibilities in detail

Create users. The admin creates accounts for any role directly from the dashboard, including other administrators. Creating a collector automatically creates the collector profile (vehicle, area); creating a recycler automatically creates the company profile (company name, location, contact person).

Manage collectors. Collectors appear in the assignment dropdown and in the **collector workload** line, which counts each collector's open jobs so work can be balanced.

Manage recyclers. Recycling partners are listed with company name, site location, contact person, and phone. They populate the collector's transfer dropdown.

Assign collectors. For each request row the admin selects a collector and presses **Assign**. Reassignment is allowed: assigning a different collector to an already-assigned request updates the assignment and clears the previous acceptance.

Monitor requests. Six live counters sit above the request table: today's requests, pending assignment, assigned and in progress, waiting at the recycler, completed, and failed. The table supports free-text search across location, citizen name and battery type, status filtering, newest-first sorting, and paging at ten rows per page.

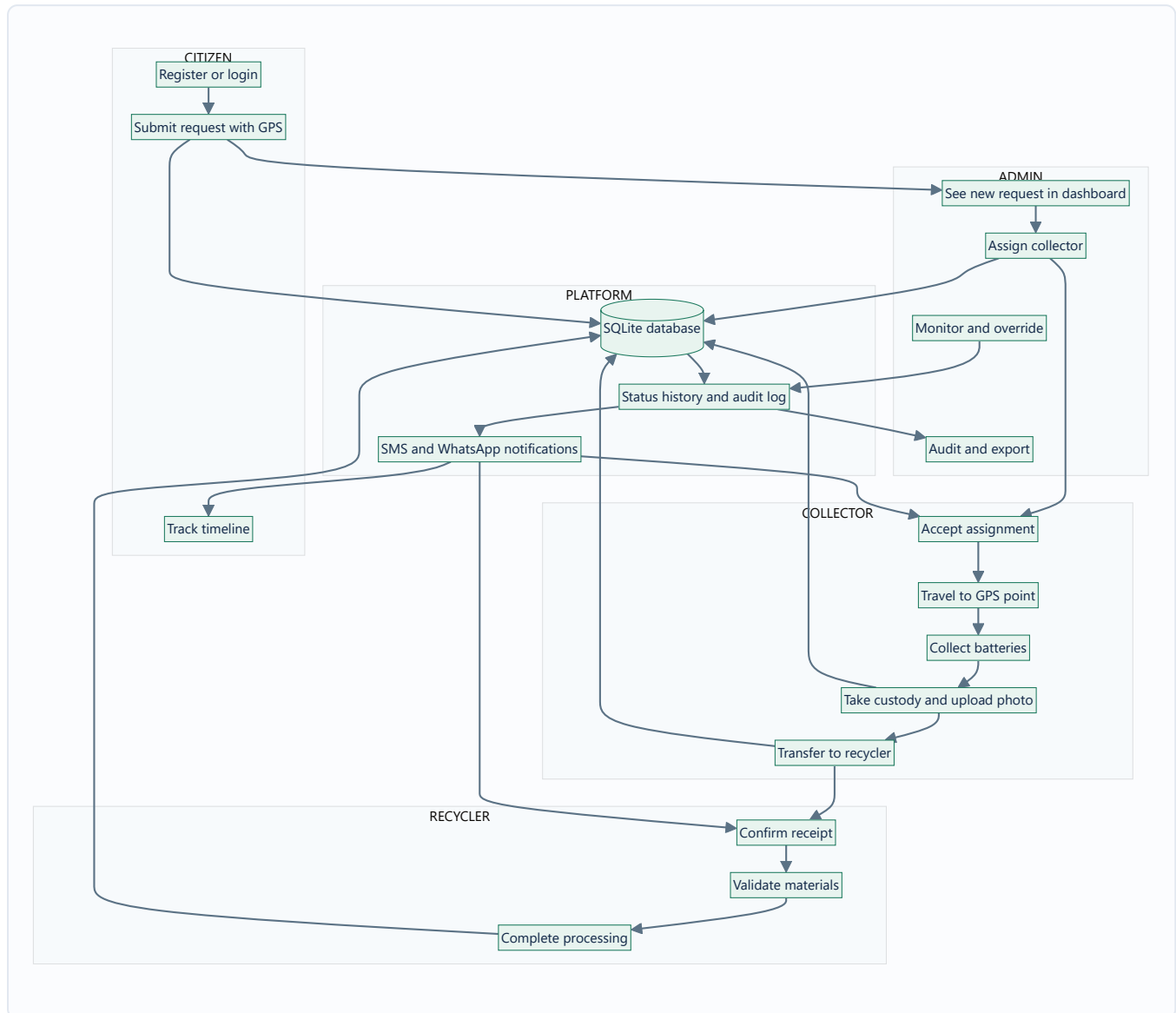
Override statuses. The admin can force any request to any status. This is intended for exceptions — a collector whose phone died, a load recovered after a failed transfer. Overrides bypass the transition rules but **never** bypass the audit trail: the override flag, the actor, the old status, and the new status are all recorded.

Review audit logs. A dedicated screen lists actions newest-first with timestamp, action type, the person responsible, and the affected record.

Generate reports. Export CSV downloads every request with citizen, location, battery type, quantity, status, and creation time — ready for Excel or for a supervisor's report.

6. End-to-End Flowchart

This is the complete operational picture across all four roles.



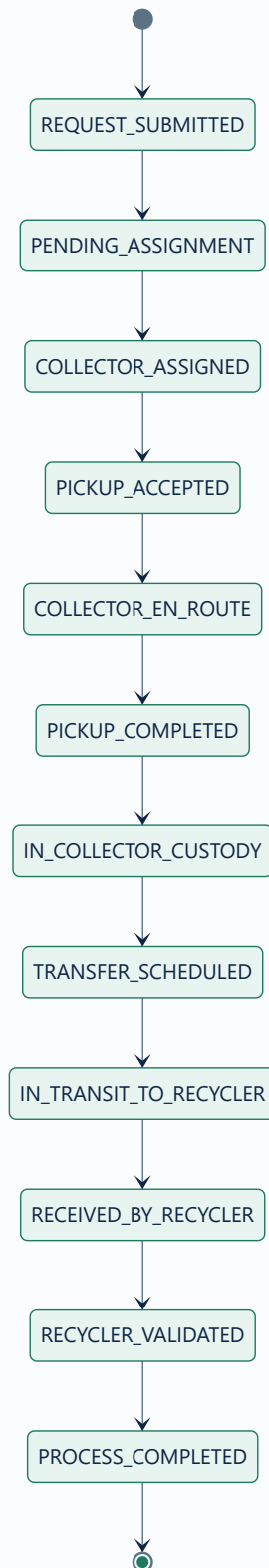
6.1 The six operational phases

Phase	Trigger	Owner	Result
1. Request submission	Citizen submits the form	Citizen	Request stored, admins alerted
2. Assignment	Admin picks a collector	Admin	Named collector, collector alerted
3. Collector pickup	Collector accepts and travels	Collector	Batteries physically collected
4. Custody transfer	Collector selects a recycler	Collector	Transfer record created, load in transit
5. Recycler acceptance	Delivery arrives	Recycler	Receipt confirmed and materials validated
6. Completion	Processing intake finished	Recycler	Terminal status, citizen notified

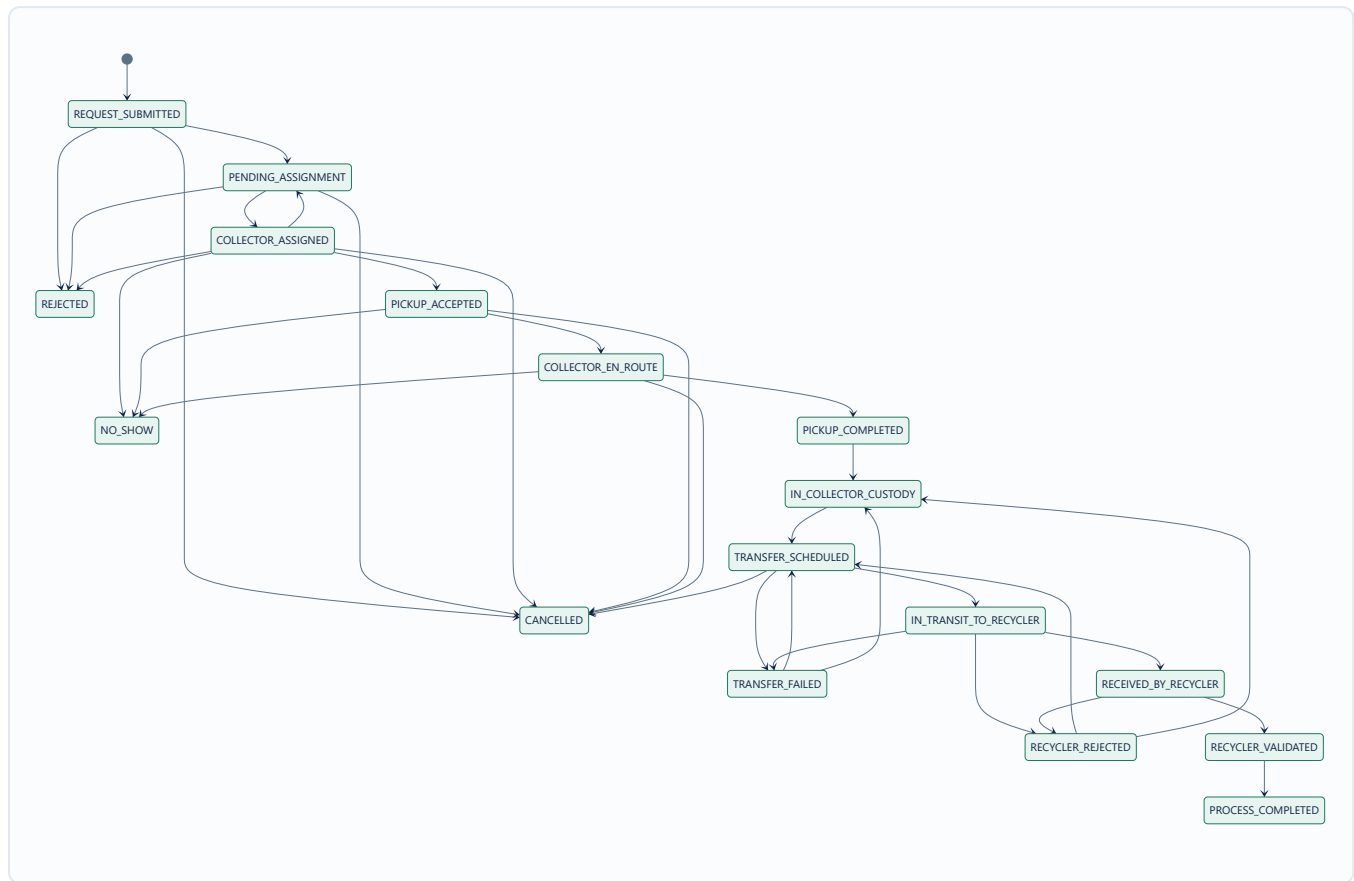
7. Status Lifecycle Diagram

The lifecycle is the heart of the system. The server refuses any move that is not on this map.

7.1 Success path



7.2 Full lifecycle including failure paths



7.3 What each status means in plain language

Status	Business meaning	Who normally causes it
REQUEST_SUBMITTED	The citizen has raised a request; nothing has been done yet	Citizen
PENDING_ASSIGNMENT	In the operations queue, waiting for a collector	System, automatically
COLLECTOR_ASSIGNED	A named collector has been given the job	Admin
PICKUP_ACCEPTED	The collector has agreed to do it	Collector
COLLECTOR_EN_ROUTE	The collector is travelling to the address	Collector
PICKUP_COMPLETED	The batteries have been handed over at the door	Collector
IN_COLLECTOR_CUSTODY	The collector is now formally holding the material	Collector
TRANSFER_SCHEDULED	A recycler has been chosen for this load	Collector
IN_TRANSIT_TO_RECYCLER	The load is on the road to the recycling site	Collector
RECEIVED_BY_RECYCLER	The recycler physically has the load	Recycler
RECYCLER_VALIDATED	The materials have been checked and accepted	Recycler
PROCESS_COMPLETED	The job is finished and closed	Recycler

7.4 Failure states

Status	Business meaning	Recovery
CANCELLED	The citizen or admin called the request off	Terminal; raise a new request

Status	Business meaning	Recovery
REJECTED	Operations declined the request (out of area, unsafe, invalid)	Terminal; raise a new request
NO_SHOW	The collector arrived and there was nothing to collect	Admin reassigns or cancels
TRANSFER_FAILED	The delivery to the recycler did not succeed	Back to TRANSFER_SCHEDULED or collector custody
RECYCLER_REJECTED	The recycler refused the load on inspection	Back to TRANSFER_SCHEDULED or collector custody

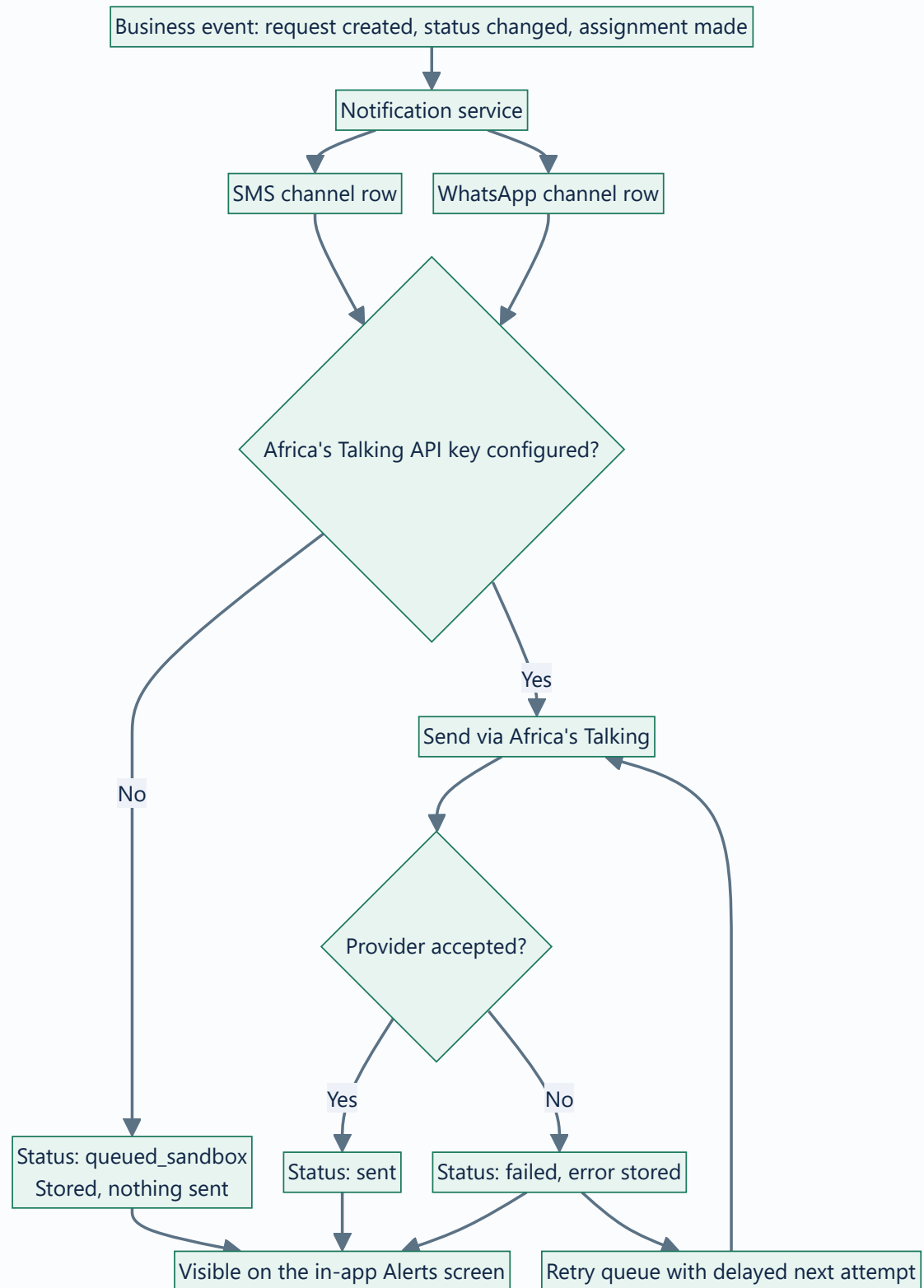
7.5 What happens on every single transition

This is guaranteed by the server, for every status change including admin overrides:

1. **The request record is updated** with the new status and a fresh "last updated" timestamp.
2. **A timeline entry is written** to status history: old status, new status, the user who did it, the exact UTC time, and a note.
3. **An audit record is written** capturing the action, the actor, the record affected, and whether an override was used.
4. **Notifications are generated** for the citizen who owns the request and for the assigned collector, on both SMS and WhatsApp channels.
5. **The change becomes immediately visible** to every role that is allowed to see that request — there is no cache to refresh and no separate copy of the data.

8. Notification Flow

8.1 How a notification is produced



Every message is written to the database **before** any send is attempted. That means the alert history is complete and auditable even when no SMS provider is connected — which is exactly the state of the POC today.

8.2 SMS flow

The SMS channel targets **Africa's Talking**, the preferred vendor for Tanzania. Each message stores the recipient phone number, the linked user, the message text, the provider name, the delivery status, the attempt count, and any error text. With no API key configured the message is stored with status `queued_sandbox` and nothing leaves the machine. Adding a live key in the environment file switches the same code path to real delivery without any change to the business logic.

8.3 WhatsApp flow

The WhatsApp channel is a parallel adapter behind the same notification interface, also aimed at Africa's Talking. It writes its own row for the same event, so the delivery record for SMS and WhatsApp is tracked separately. In the POC live WhatsApp sending is deliberately disabled; the adapter records acceptance and reserves the endpoint.

8.4 Who is alerted, and when

Event	Citizen	Collector	Recycler	Admin
Account created	Welcome message	Welcome message	Welcome message	—
Pickup request submitted	Confirmation with request number	—	—	Every admin: "new pickup needs assignment"
Collector assigned	Status change alert	"You were assigned pickup #N at [location]"	—	—
Collector accepts / travels / collects	Status change alert	Status change alert	—	—
Custody transfer started	Status change alert	Status change alert	"Incoming transfer for request #N"	—
Recycler confirms / validates / completes	Status change alert	Status change alert	—	—
Any admin override	Status change alert	Status change alert	—	—
Password reset requested	Reset code by SMS and WhatsApp	Same	Same	Same

All recipients can also read their full alert history in the application under **Alerts**, regardless of whether the provider delivered.

9. Business Rules

These are the rules the software actually enforces. They are grouped by area.

9.1 Identity and access

#	Rule
BR-01	Phone numbers must match the Tanzanian format <code>255XXXXXXXX</code> (twelve digits starting 255).
BR-02	Passwords must be at least eight characters.
BR-03	Email addresses and phone numbers are unique; duplicates are refused.
BR-04	Self-registration can only create citizen, collector, or recycler accounts. Administrator accounts can only be created by an existing administrator.
BR-05	Passwords are stored only as salted hashes and are never returned by any endpoint.
BR-06	Login sessions expire after eight hours and must be renewed by signing in again.
BR-07	Deactivated accounts (<code>active_flag = 0</code>) cannot log in or use a previously issued token.
BR-08	Password reset codes expire thirty minutes after they are issued and can be used once.
BR-09	Password reset never reveals whether a phone number exists in the system.

9.2 Pickup requests

#	Rule
BR-10	Only citizens (and administrators acting on their behalf) can create pickup requests.
BR-11	Address, battery type, and a quantity of at least one are mandatory.
BR-12	A new request is created as <code>REQUEST_SUBMITTED</code> and is advanced to <code>PENDING_ASSIGNMENT</code> automatically; both steps appear in the timeline.
BR-13	GPS coordinates are optional but strongly encouraged; when present they generate navigation links.
BR-14	A citizen can only see and open their own requests.

9.3 Assignment and custody

#	Rule
BR-15	Only an administrator can assign a collector to a request.

#	Rule
BR-16	A request has at most one active assignment; reassigning replaces the collector and clears the previous acceptance.
BR-17	A collector can only accept a request that is assigned to them.
BR-18	A collector can only hand over a request they currently hold.
BR-19	A handover must name a recycling partner; it creates a custody transfer record with the collector, recycler, and date.
BR-20	Custody passes to the recycler only when the recycler confirms receipt — not when the vehicle leaves.

9.4 Lifecycle integrity

#	Rule
BR-21	Status transitions must follow the allowed map in section 7.2; anything else is refused with a conflict error.
BR-22	Only administrators may override the transition rules, and only deliberately.
BR-23	Overrides are recorded as overrides in the audit trail.
BR-24	Re-applying the current status is harmless and does not create a duplicate timeline entry.
BR-25	Every transition writes timeline, audit, timestamp, actor, and notifications — with no exception.
BR-26	<code>PROCESS_COMPLETED</code> , <code>CANCELLED</code> , and <code>REJECTED</code> are terminal in normal operation.

9.5 Evidence, reporting, and platform

#	Rule
BR-27	Evidence photos are limited to five megabytes and are stored per request, with the uploader and timestamp recorded.
BR-28	Citizens do not upload evidence; collectors and recyclers do.
BR-29	Audit logs and the CSV export are administrator-only.
BR-30	The dashboard shows ten requests per page, newest first.
BR-31	Repeated requests from the same address are rate-limited to protect the service.
BR-32	The M-Pesa connector is sandbox-only and cannot move real money; every sandbox transaction is still recorded and audited.
BR-33	The database is the only source of truth; nothing that matters is kept in the browser.

10. Operational Workflow Summary

Written for management. No technical vocabulary.

A resident in Dar es Salaam has four dead car batteries in their yard. They open VoltRescue on their phone, create an account with their phone number, and fill one short form: where they are, what kind of batteries, how many. They drop a pin on the map, or let the phone find them. They press **Request pickup**.

That request lands instantly in the operations queue. Every administrator gets an alert saying a new pickup is waiting. On the operations dashboard the request appears at the top of the table, marked *pending assignment*, and the counter for pending work goes up by one.

The administrator looks at the collector workload line — Juma has three open jobs, Neema has one — and assigns Neema. Neema's phone gets a message: *you were assigned pickup #12 at Kariakoo*. The resident also gets a message telling them a collector has been assigned.

Neema opens her assignments list, presses **Accept**, and the resident is told the collector accepted. When she sets off she presses **En route**, and the resident sees that too. She taps **Navigate** and her map app opens directions straight to the pin the resident dropped.

At the gate she counts the batteries, loads them, presses **Pickup completed**, takes a photo of the load as evidence, and presses **In collector custody**. From that moment the system says, in writing, that Neema is holding four lead-acid batteries from that address.

Back at the yard she chooses GreenCycle Dar from the recycler list and presses **Transfer to recycler**. The system creates a custody transfer record and marks the load as in transit. GreenCycle receives an alert that a load is coming.

When the vehicle arrives at Vingunguti, GreenCycle's staff open their screen, see the inbound load, and press **Confirm receipt**. Custody has now formally moved from Neema to GreenCycle. They weigh and sort the batteries and press **Validate materials**, then after processing intake they press **Complete processing**.

The resident gets a final message: the job is done. The whole journey — twelve stages, every timestamp, every person, every photo — is stored permanently. If a supervisor asks in six months who held those batteries on that day, the answer is one click away in the audit log, and the whole month's activity can be exported to a spreadsheet in one press.

If something goes wrong, the system has a place for it. Nobody at the door? *No-show*, and the administrator reassigns. Delivery failed? *Transfer failed*, and the load goes back into collector custody. Recycler refuses the load? *Recycler rejected*, and it is rescheduled. A collector's phone died mid-job? The administrator overrides the status manually — and that override is itself recorded, with their name against it.

What this POC does not yet do: it does not photograph a battery and tell you what it is worth, it does not pay anybody, it does not issue recycling certificates, and it does not calculate carbon savings. Those are Phase 2. What it does do is prove the operational backbone — request, assign, collect, transfer, receive, complete — works end to end with real data, real roles, and a complete audit trail.

End of Document 1. For step-by-step usage instructions see [02_VoltRescue_User_Guide.md](#). For technical implementation see [03_VoltRescue_Technical_Design_Handbook.md](#). For the readiness assessment and roadmap see [04_VoltRescue_Final_Readiness_Report.md](#).